

CSC00004 – Introduction to Information Technology

Session 04

Future-Ready Learning with GenAI Skills

Presenter:

XXX



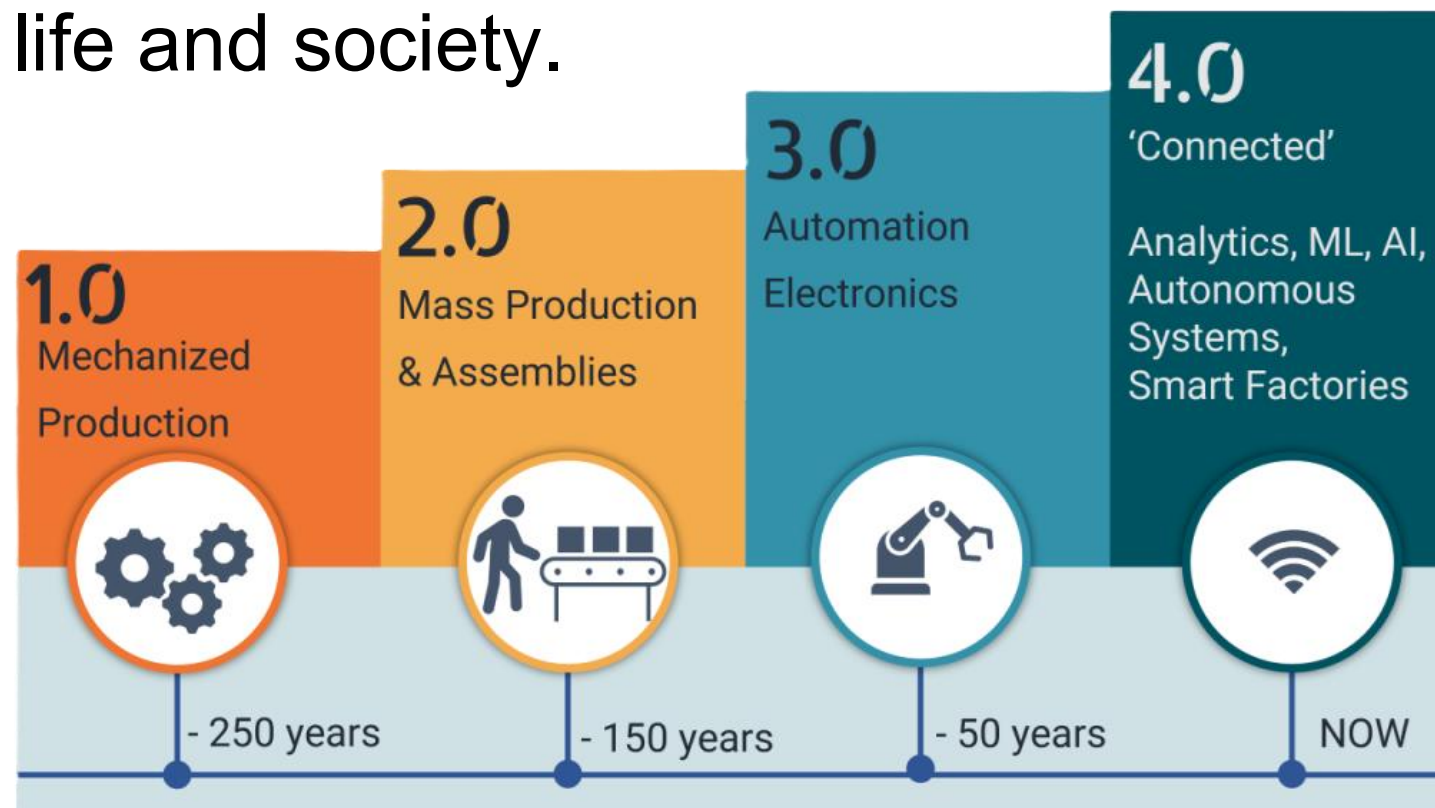
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- 1 Introduction to GenAI
- 2 Prompting: Effective Communication with AI
- 3 GenAI skills in learning and research efficiency
- 4 GenAI awareness and Human response

1. Introduction to AI & GenAI

Introduction

The Fourth Industrial Revolution is developing rapidly, with Artificial Intelligence (AI) exerting a deep and wide-reaching influence across all areas of life and society.





AI as a Global Megatrend

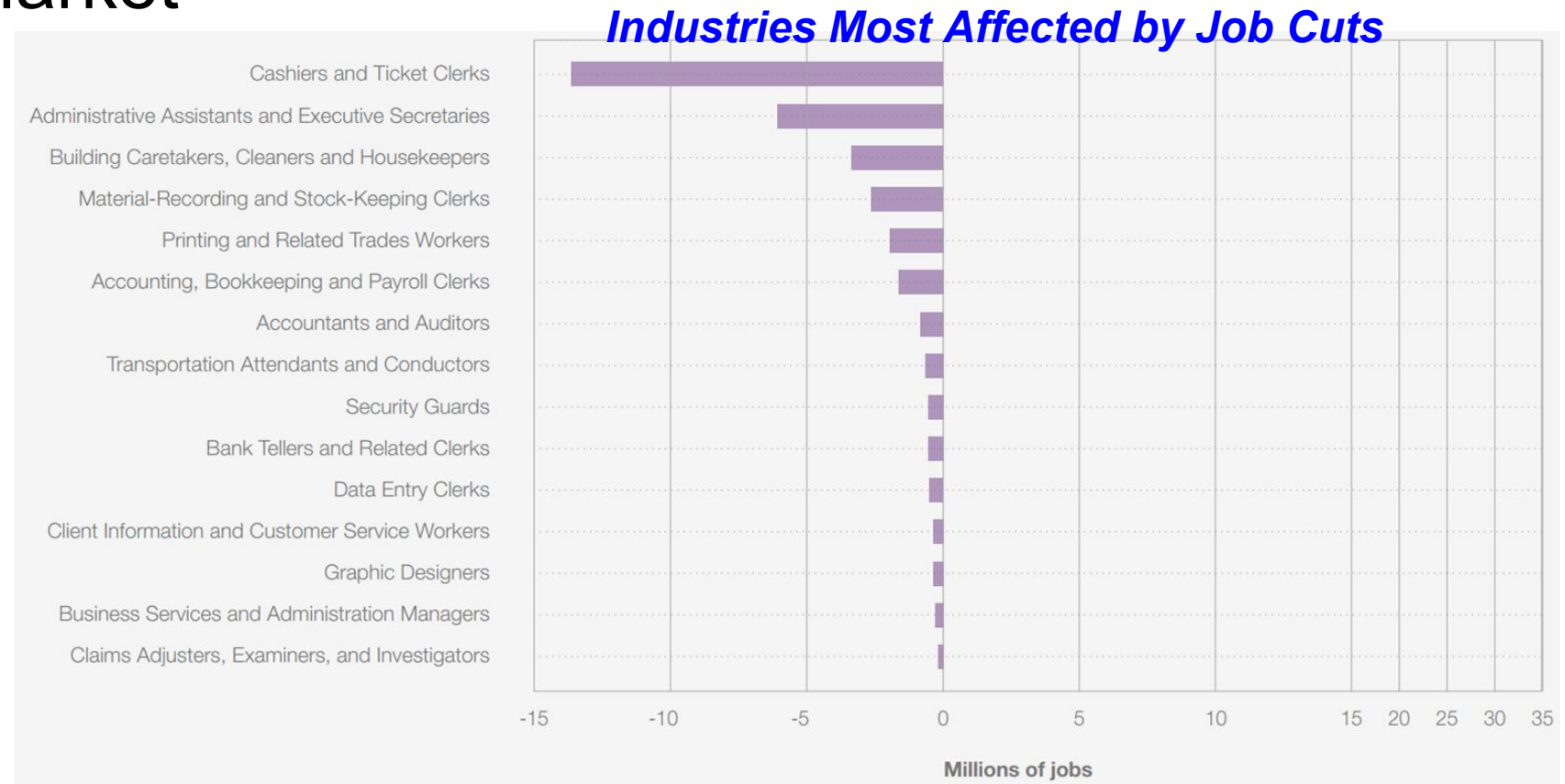
Explosive Growth in the AI Market

Expanding Applications Across All Sectors

Emergence of New Jobs and Industries

Introduction

The Far-Reaching Impact of AI and Its Applications on the Future Job Market



Source: *The Future of Jobs Report*, WEF, 2025

Introduction

Essential Skills for the New Age



Technical Skills

Proficiency in AI and digital tools



Creative Thinking

Innovation and problem-solving



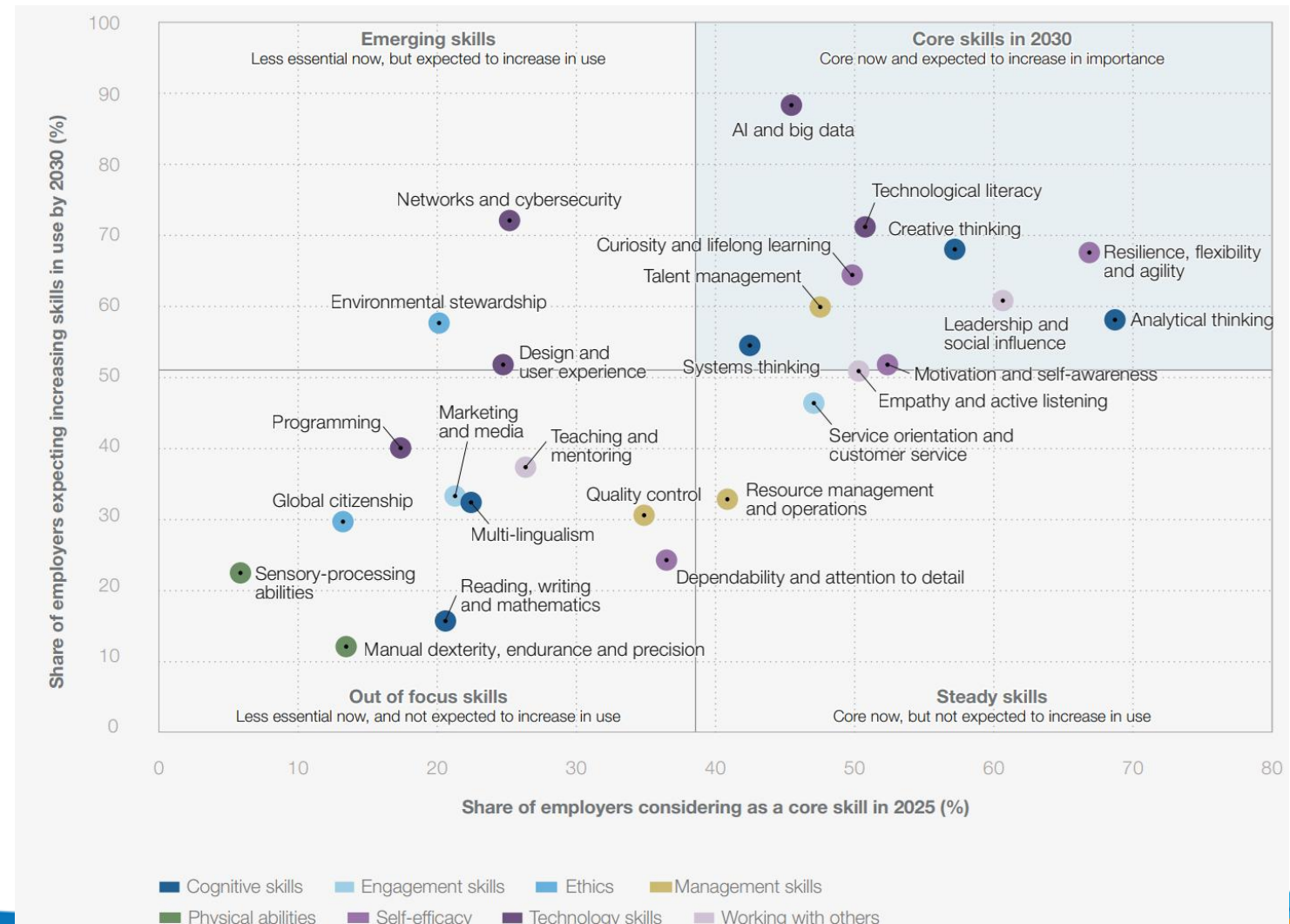
Emotional Intelligence

Human-centric skills



Lifelong Learning

Adaptability and continuous education



Source: The Future of Jobs Report, WEF, 2025

Introduction

GenAI in Interior Decorating/ Design



<https://quantrimang.com/cong-nghe/ung-dung-thiet-ke-noi-that-ho-tro-ai-202237>

Introduction

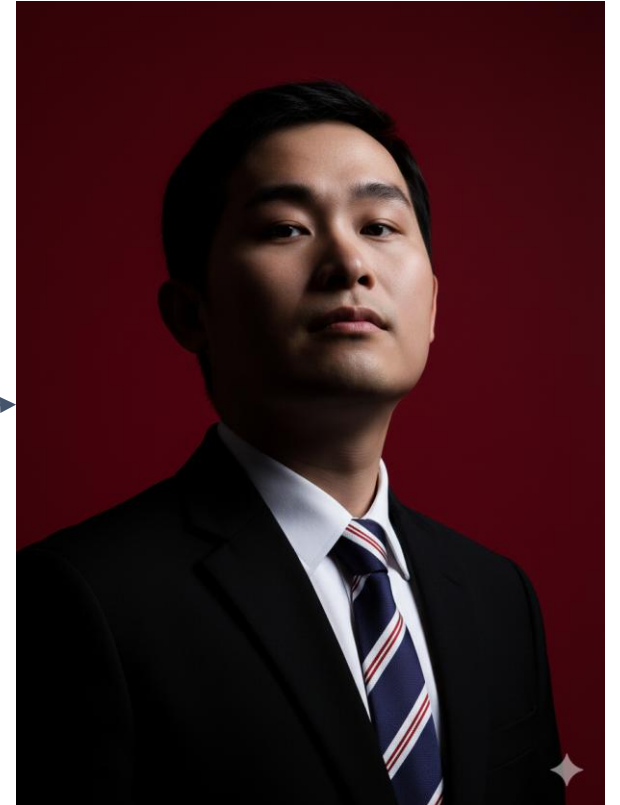
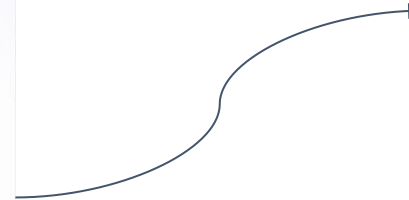
GenAI in Food Decoration/ Advertisement



<https://ai.google.dev/gemini-api/docs/image-generation?hl=vi>

Introduction

GenAI in Profile Image Generation



Introduction

GenAI in Music/ Video/ Image



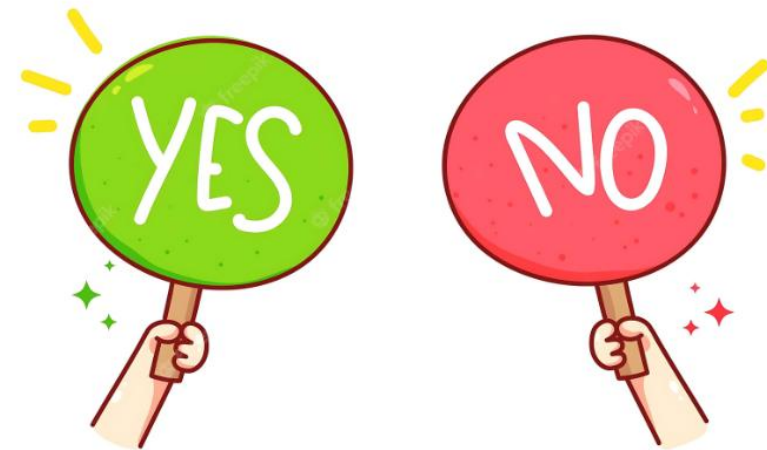
Singer - Performer: GenAI



Singer - Performer: Human

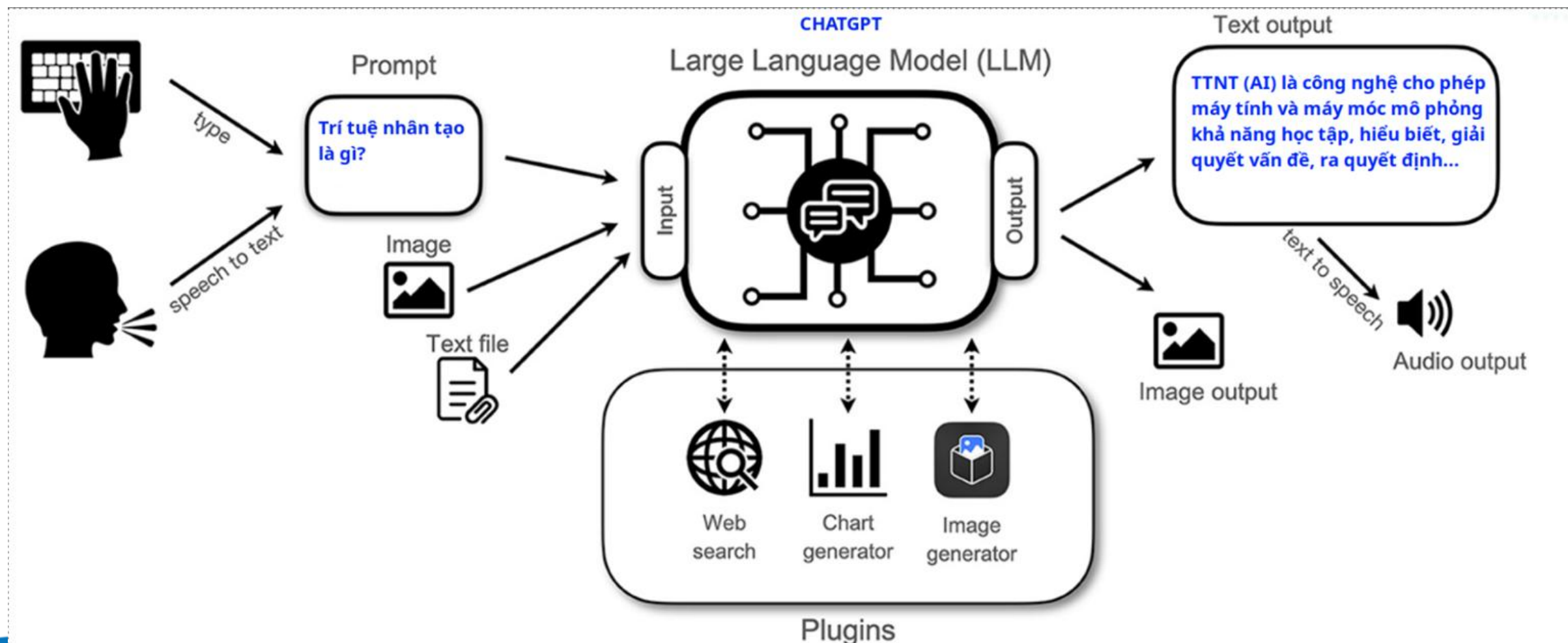
Introduction

Have you ever thought that
computers can do everything?



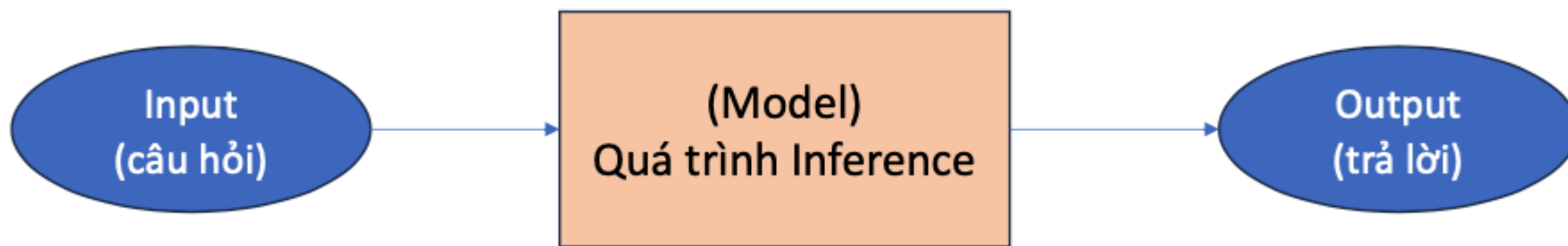
2. Prompting: Effective Communication with AI

GenAI flowchart

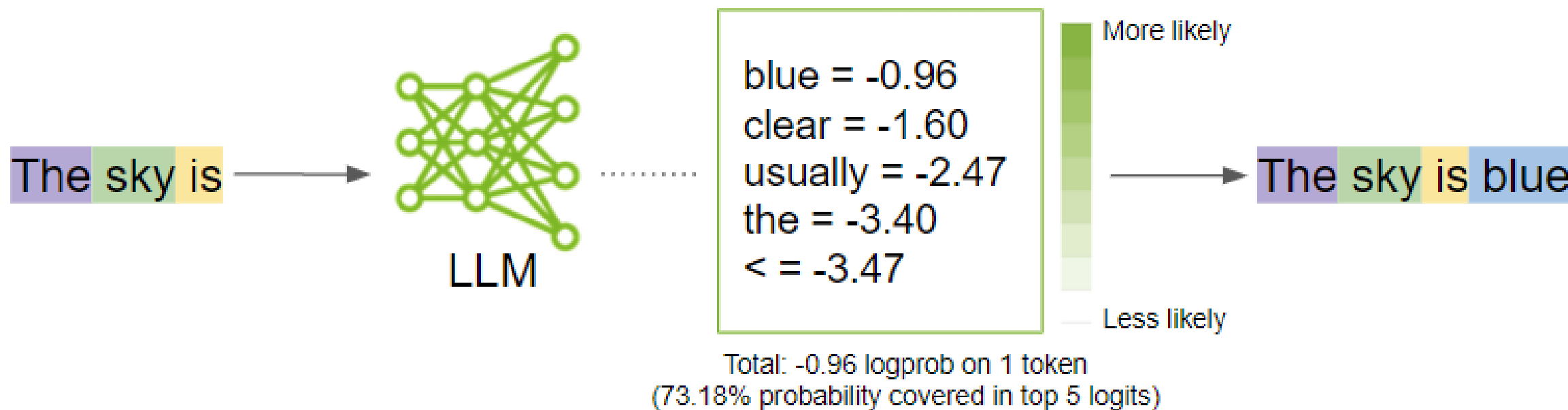


How GenAI work

Gen AI analyzes your **question** and **predicts** the most likely **output** based on **model** it learned from vast amounts of text data



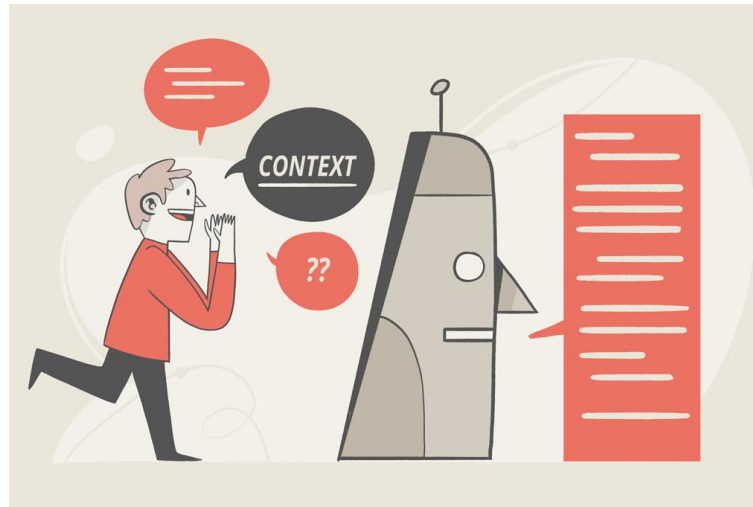
How GenAI work



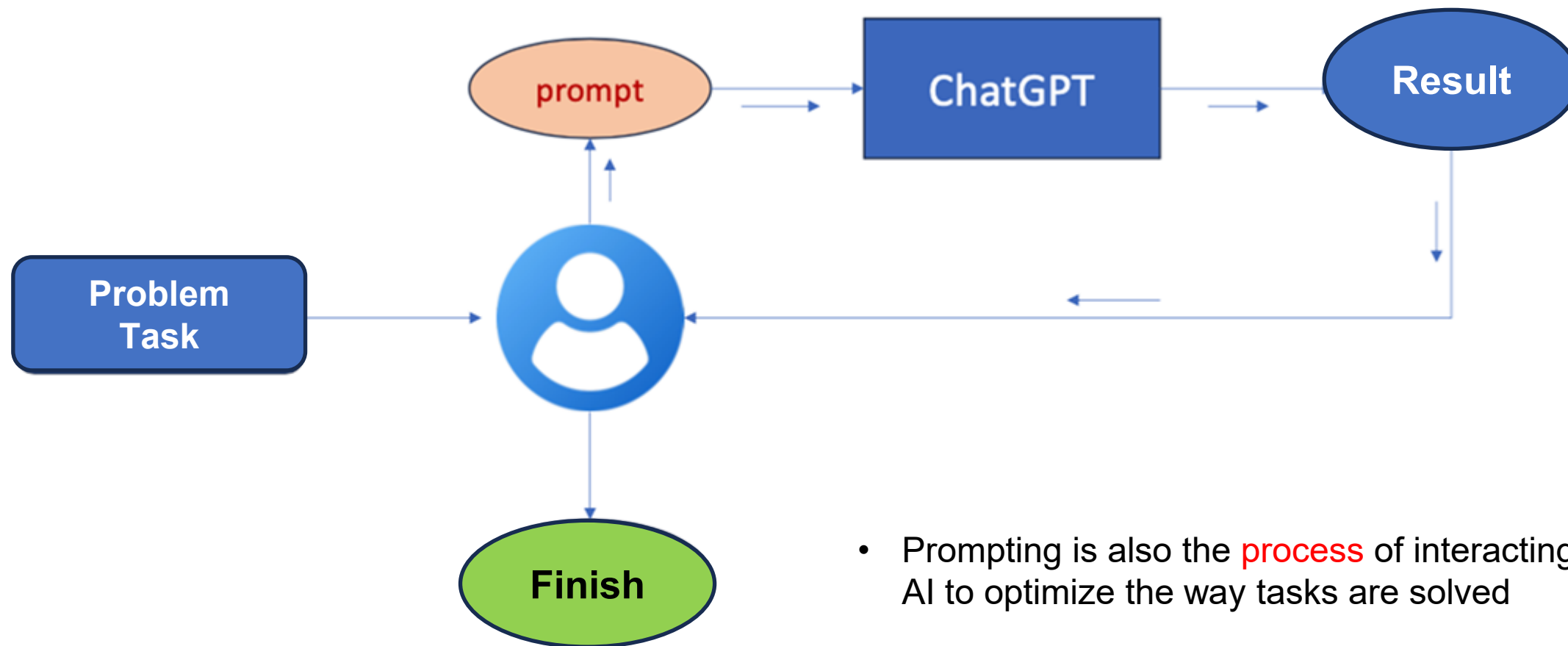
What are Prompt & Prompting?

Prompt: Instructions that guide GenAI responses

Prompting: Techniques for precise descriptions to achieve desired results



What are Prompt & Prompting?



- Prompting is also the **process** of interacting with AI to optimize the way tasks are solved

Why is Prompting Important?

Direct requests:

- "What is GenAI?"
- "Summarize this document"



Suitable for:

- General knowledge questions
- Clear and simple tasks

Direct requests:

- "Write an email to schedule a meeting with a teacher"
- "Plan a teambuilding event"



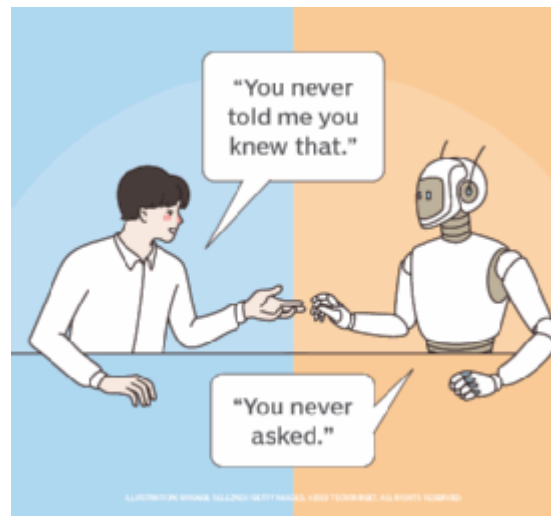
Not suitable for:

- Tasks with specific audience, context, and objectives
- Complex tasks that need to be broken down for processing
- Tasks requiring investigation, analysis, and data synthesis
- Tasks requiring multi-step reasoning processes...

Why is Prompting Important?

Good Prompt → AI responds accurately and as requested

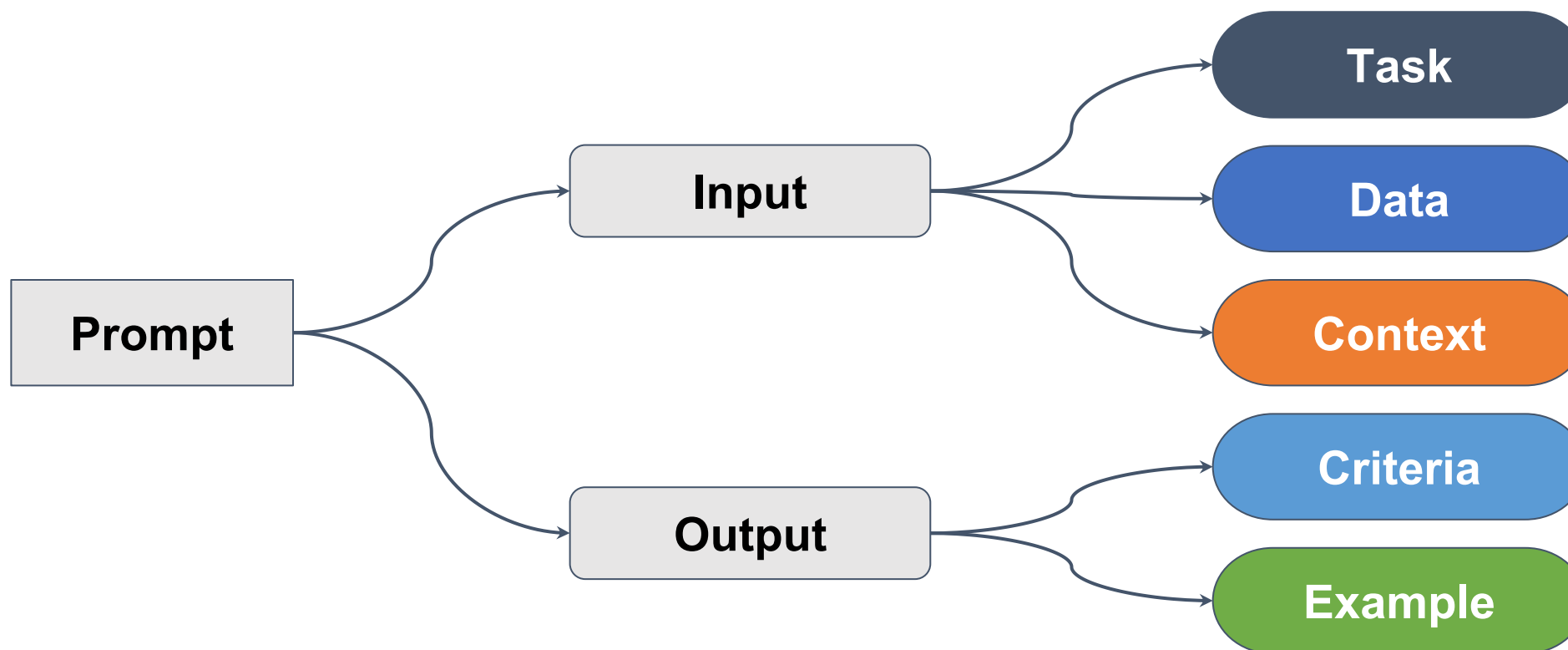
Unclear Prompt → Poor quality, rambling results



Prompt structure



Prompt Structure



Prompt Structure

Task

Multimedia Creation
Create images, graphics,
audio and video

Content Creation
Create new or continue
writing content

Data Analysis
Analyze and predict
trends in data

Content Analysis
Summarize and analyze
existing content

Question Processing
Answer questions or ask
questions back

Content Translation
Translate content between
different languages



Prompt Structure

Task

Prompt example

Write a blog post about summer skincare

Analyze the sentiment of this review

Draw a cat wearing a cowboy hat in a desert setting

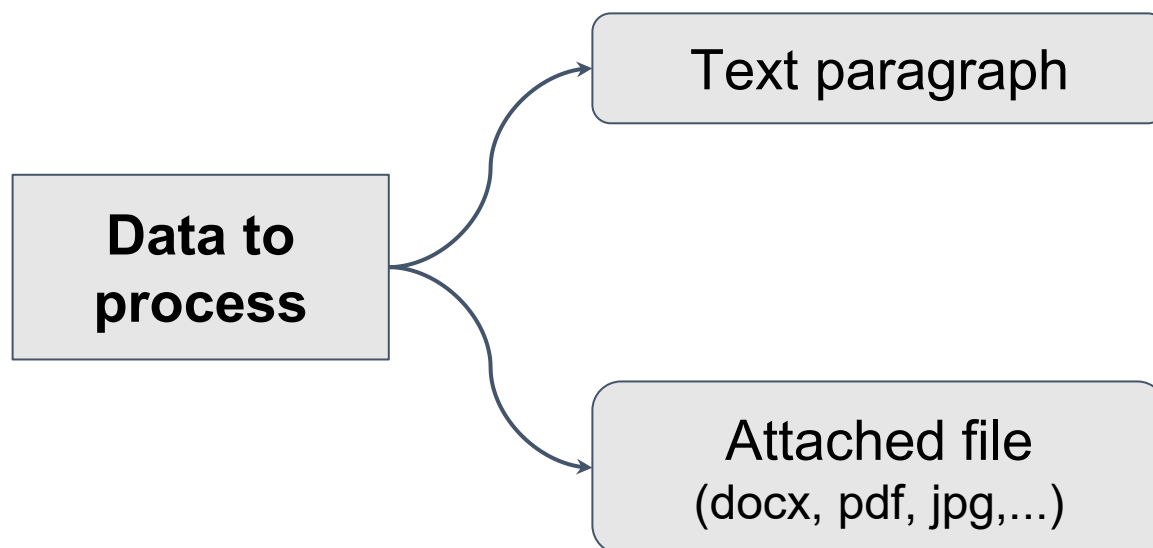
Predict sales trends for next month

Ask me some questions about GenAI

Translate this paragraph from English to Vietnamese

Prompt Structure

Data



Translate this paragraph from English to Vietnamese
GenAI (Generative AI) is a subset of artificial intelligence focused on creating new, original content rather than just analyzing or classifying existing data...

Predict sales trends for next month based on the attached report

Convert this image to Ghibli style

Prompt Structure

Data

Context Window: The maximum amount of information that AI can "remember" and process simultaneously in one response

- If too much content is entered, AI will forget what's at the beginning
When working with long text:
 - Break content into parts to process separately
 - Summarize main content first for AI to remember longer
 - Remind context if needed (e.g., "earlier I mentioned content A")

Prompt Structure

Context

AI knows how to do it but doesn't know:

- The **role** of the executor
- The **context** of the task
- The structure of **data** to process



Provide AI with task context:

- **Role**
Profession, knowledge, experience,... of the executor
- **Context**
What specific situation does this task occur in?
- **Data**
Detailed description of data to process if necessary

Prompt Structure

Context

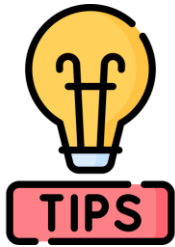
You are a digital content specialist (role). I am about to launch a new product specifically for Gen Z (context). Please help me write a short advertisement.

You are a translation expert (role). I need to translate an academic document from English to Vietnamese for use in educational research (context). Please help me translate the passage below.

You are a professional administrative assistant (role). I just received an email from a partner complaining about delays in project A (context). Please help me write an apology email, explain the reasons, and commit to a new completion time.

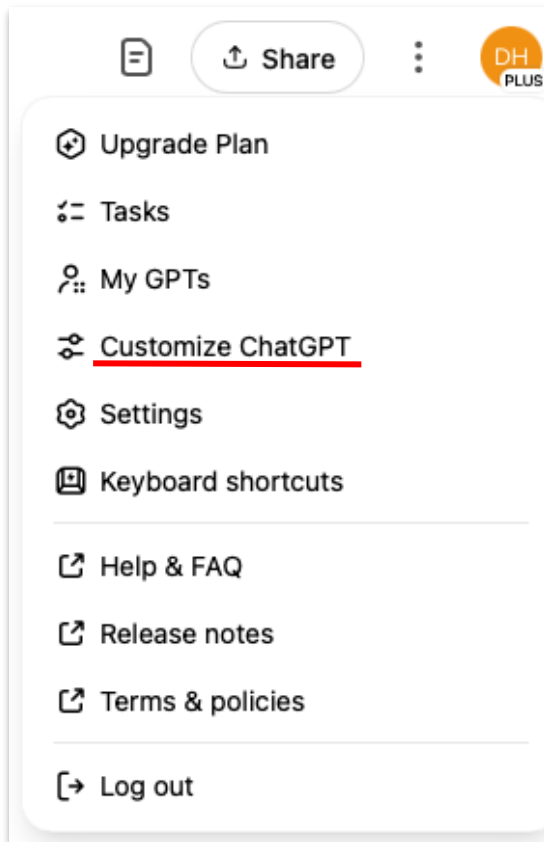
You are a data analysis expert (role). I have an Excel file containing sales data for the first 6 months of the year with columns: time, region, revenue, number of orders, discount (data). Please help me analyze the data to find: (1) monthly revenue trends, (2) region with highest revenue, (3) average effectiveness of discount programs.

Prompt Structure



Customize ChatGPT

Context



Customize ChatGPT

Introduce yourself to get better, more personalized responses ⓘ

What should ChatGPT call you?

Nickname

What do you do?

Level 10 mage

What traits should ChatGPT have? ⓘ

Describe or select traits

+ Chatty

+ Witty

+ Straight shooting

+ Encouraging

+ Gen Z

+ Skeptical

+ Traditional

+ Forward thinking

+ Poetic



Anything else ChatGPT should know about you? ⓘ

Interests, values, or preferences to keep in mind

Advanced ▾

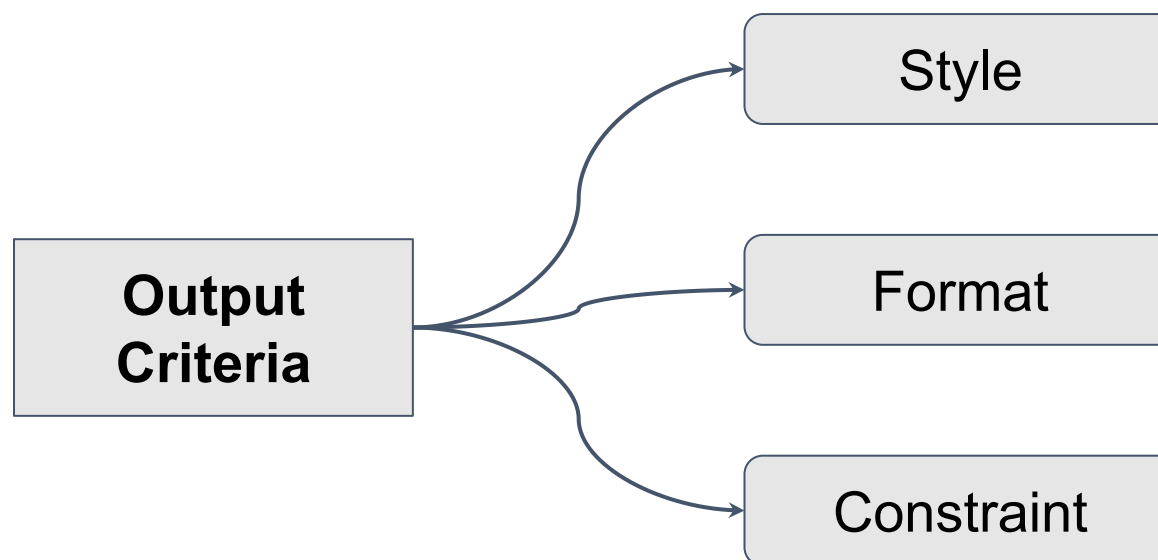
☒ Enable for new chats

Cancel

Save

Prompt Structure

Criteria



Prompt Structure

Criteria

Style

Style, tone

- Write an analytical report on the impact of inflation on Vietnam's economy, **using an analytical style.**
- Write an introduction about a famous tourist destination **with a descriptive and inviting tone.**
- Write a seminar invitation letter **with a formal style.**
- Write an essay about the importance of language in connecting people **with a profound and inspiring tone.**
- Write a product description for a new t-shirt **with a humorous style.**
- Write a paragraph about the beauty of language **with a creative and poetic tone....**

Prompt Structure

Criteria

Format

Định dạng đầu ra

 Desired format	 Suggested prompt	 Usage notes
Table	<i>Present results in a table format with columns: Category – Status – Time.</i>	Useful for statistics, comparisons, data synthesis.
Bullet points	<i>Summarize the following content into 5 bullet points, each no more than 12 words.</i>	Suitable for emails, slides, brief reports.
Word (.docx)	<i>Write me a guide document in Word format, including title, sections, and clear formatting.</i>	Easy to copy into MS Word or Google Docs.
PowerPoint (.pptx)	<i>Create content for a presentation with 5 slides, each slide with a title and concise content.</i>	Content can be copied directly into PowerPoint.

Prompt Structure

Criteria

Format

Định dạng đầu ra

 Desired format	 Suggested prompt	 Usage notes
PDF	<i>Create a brief course summary for me to convert to PDF file.</i>	AI cannot directly create PDFs, but can create content to save.
Mind map	<i>Present the following content in a mind map format, with the main topic in the center and sub-branches around it.</i>	Can be used in software like XMind, MindMeister.
Presentation slides	<i>Create an outline for a 30-minute lecture about ChatGPT, including slide titles and suggested content for each section.</i>	Suitable for students preparing content.
Images	<i>Create an illustration for the topic 'AI in education', suitable for inserting into slides or documents.</i>	Images can be inserted into Word/PowerPoint.

Prompt Structure

Criteria

Constraint

Limit output + Target audience

- Write an analysis paragraph on the impact of inflation on Vietnam's economy. **Limit to 150 words, use recent data if available.**
- Write a short report **(100-150 words)** on Vietnam's economic situation in 2025, **targeting business executives.**
- Explain the concept of GenAI **to an economics student.**
- Write a paragraph encouraging foreign language learning, **targeting first-year students.**
- Write a short description **(maximum 100 words)** about Ha Long Bay for use in tourism promotional materials **for international visitors.**
- List 3 benefits of e-commerce for the economy, **each benefit no more than 20 words.**

Prompt Structure

Criteria

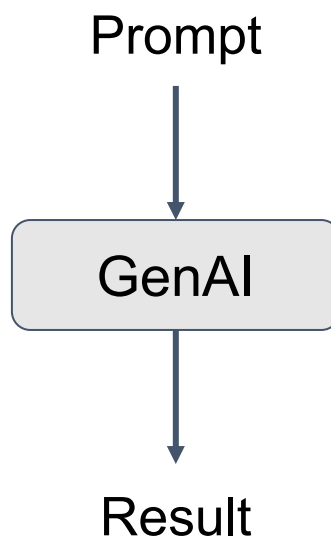
You are a digital content specialist (role). I am about to launch a new product specifically for Gen Z (context). Please help me write a short advertisement with a humorous tone, including emojis, length no more than 100 words (criteria).

You are a professional administrative assistant (role). I just received an email from a partner complaining about delays in project A (context). Please help me write an apology email, explain the reasons, and commit to a new completion time. Friendly but clear tone, not overly defensive. Output format is a complete email with formal greeting and closing (criteria).

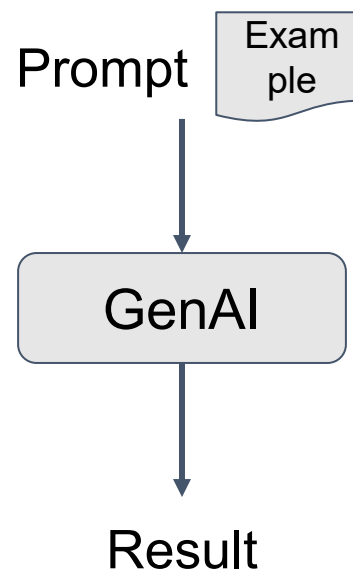
You are a data analysis expert (role). I have an Excel file containing sales data for the first 6 months of the year with columns: time, region, revenue, number of orders, discount (data). Please help me analyze the data to find: (1) monthly revenue trends, (2) region with highest revenue, (3) average effectiveness of discount programs. Required analysis in natural language, use charts if possible to illustrate. Output format is an analysis summary of about 300 words, suitable for sending to management (criteria).

Prompt Structure

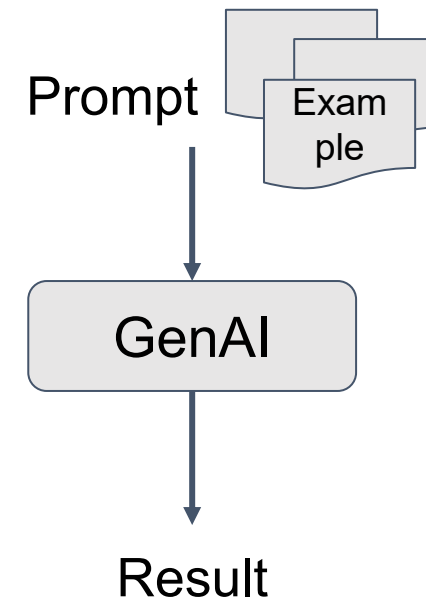
Example



Zero-shot
Prompting



One-shot
Prompting



Few-shot
Prompting

Prompt Structure

Example

Please help me write a short advertisement about t-shirts for Gen Z.

Example:

🔥 SPECIAL OFFER - LIMITED QUANTITY ✨
SUPER SALE EARLY MAY ✨ Opportunity to own
ELSA Premium 1 year at an extremely preferential
price

📁 ELSA Premium 1 Year for 1,500,000 VND, 69%
discount (Original price 4,800,000-VND) 📩 Get the
offer now

Please help me design a Facebook Feed Ad about
t-shirts for Gen Z. Reference the attached file.



One-shot
Prompting

Prompt Structure

Example

Classify the following user reviews into Positive, Negative, Neutral:

Review: "Reasonable price but delivery time is a bit slow."

Review: "I am very satisfied with the customer service."

Review: "Product is okay, not too good but not too bad either."

Example 1: Review: "This product is wonderful, good quality and fast delivery."

Classification: Positive

Example 2: Review: "Product not as expected, color different from image."

Classification: Negative

Example 3: Review: "On-time delivery but product quality is only average."

Classification: Neutral

Few-shot
Prompting



Prompt Structure

Task

Data

Context

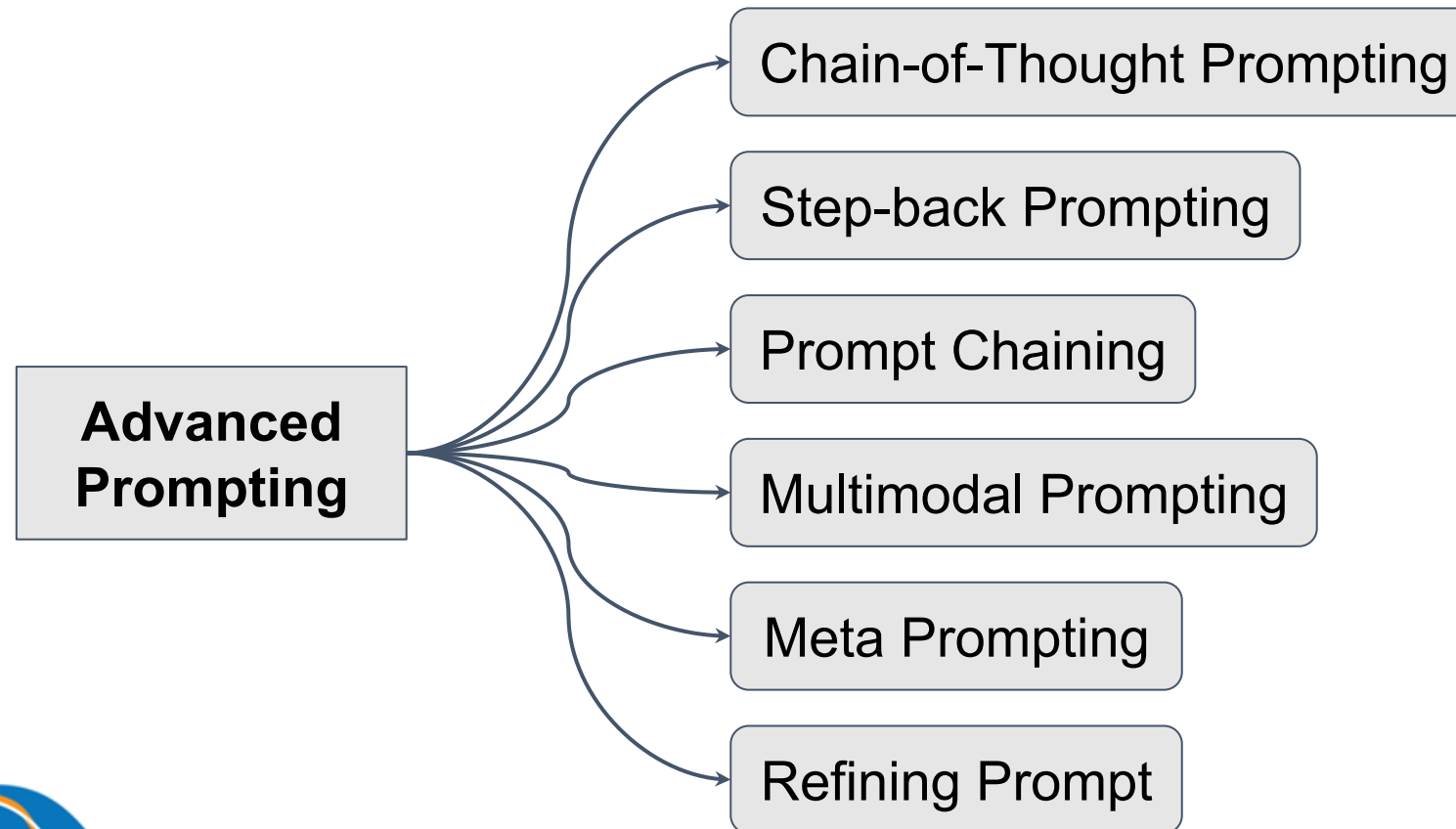
Criteria

Example

Email response prompt template

You are a professional administrative assistant. I just received an email from a partner complaining about delays in project A. Please help me write a response email that demonstrates professionalism, sincerely apologizes, provides reasonable explanation, and commits to a new completion time. Friendly but clear tone, not overly defensive. Output format is a complete email with formal greeting and closing.

Advanced Prompting



3. GenAI skills in learning and Research efficiency

Roles of AI in Learning and Research

AI does not replace research thinking, but is an intelligent assistant that helps students:

- Search and synthesize information quickly;
- Suggest research frameworks, hypotheses, or research questions;
- Analyze data (preliminary or advanced);
- Write, edit, and critique academic content.

Remember: The responsibility for critical thinking, evaluating credibility, and making final decisions **always belongs to the user.**

General method

(1) Identify the Research Question

- Determine the practical or theoretical issues that need to be understood.
- Read relevant literature to find the "research gap"
- Write a clear, specific, and measurable research question.

Example: Hallucination problem on LLM

(2) Search for Relevant Information

- Use search engines for academic resources (Google Scholar, Scopus, Web of Science...).
- Filter out articles, books, and studies directly related to the question.
- Record information sources and cite them correctly.

Example: Find key works on Hallucination, such as: C. Jeong (2024) - Fine-tuning and utilization methods of domain-specific llms, articles in JIIS,...

(3) Analyze the Information

- Read, compare, and synthesize the research results found.
- Identify trends, similarities, and differences among the studies.
- Analyze the strengths and weaknesses of each source.

Example: Strong trend across the research is the adoption of retrieval and augmentation techniques to combat hallucination

(4) Synthesize Results

- Write a literature review with a logical structure.
- Summarize the main findings from the analyzed information.
- Present the remaining research gap and directions for further research.

Example: A primary research gap is the need to move beyond evaluating only the final output's accuracy and instead prioritizing the factual soundness of the entire intermediate reasoning process.

Identify the Research Question

- The main research question is the **starting point** of every research process.
- To effectively exploit AI tools and reasoning models, we need to:
 - Rephrase the question to make it clear, specific, and focused.
 - Create sub-questions to clarify the research problem.

Identify the Research Question

Example:

- Topic: Hallucination in LLM
- Here are some sub-questions can help clarify the topic:
 - What are the root causes and underlying mechanisms that lead to hallucinations in LLMs?
 - What are the current techniques and approaches to mitigate or reduce hallucinations in LLMs?
 - What are the real-world implications and risks of LLM hallucinations in different application domains (healthcare, legal, education, etc.)?

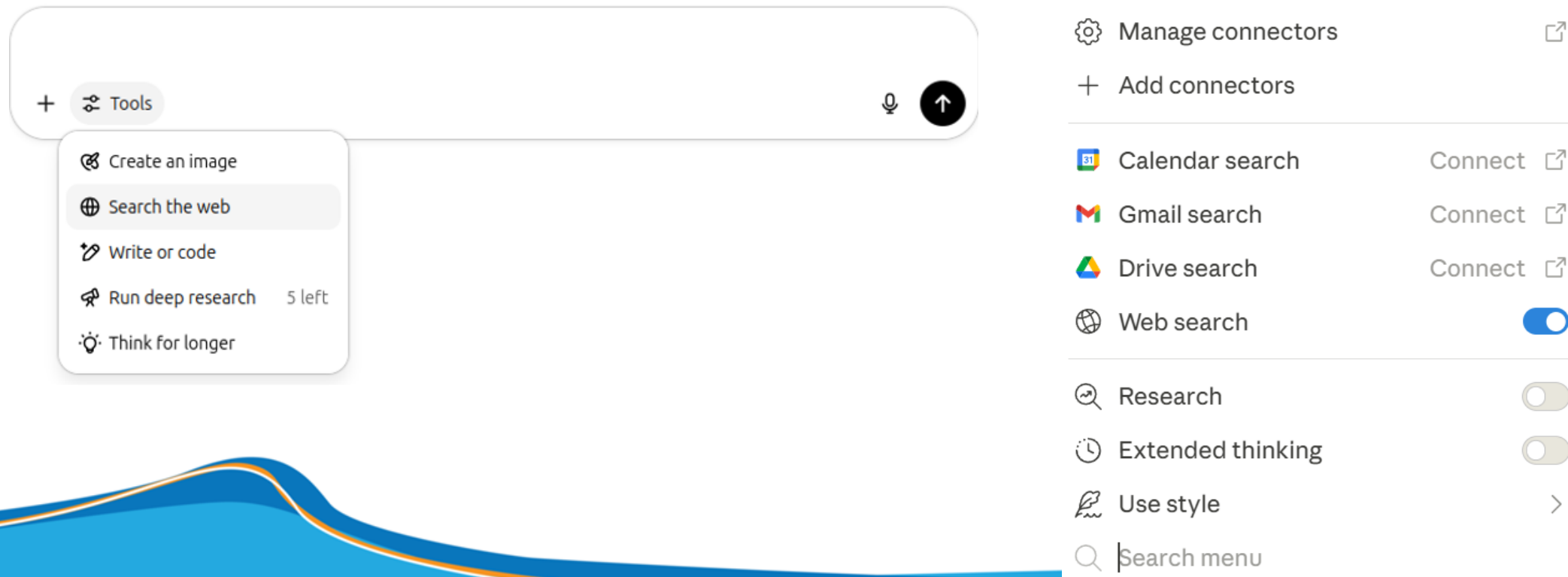
Search for Relevant Information

The importance of information searching in research:

- **Understand the research problem clearly:** Helps learners grasp the context, concepts, and core content related to the research question.
- **Identify knowledge gaps (Research gap):** Recognize what has been studied and what remains unresolved – thereby shaping new directions.
- **Build theoretical foundation and analytical framework:** Existing theories, models, or concepts will be the foundation for data analysis and argumentation.
- **Provide reliable academic evidence:** Serves as a basis for rigorous argumentation, avoiding subjectivity or unfounded speculation.

Search for Relevant Information

- Can use Google Search.
- Or LLM search (Depend on which LLM, and version)



Analyze the Information

Analyzing information is not just about reading and taking notes, but a process of **distilling knowledge to build a solid foundation for your research. Specifically, this process helps:**

- **Identify trends and similarities/differences:** Able to detect current research trends or help recognize commonalities (such as all using surveys) and differences (in subjects, contexts, conclusions...).
- **Analyze strengths and weaknesses of each document:** Helps students evaluate research quality, thereby selecting the most reliable and appropriate documents to include in the theoretical foundation or academic argumentation.

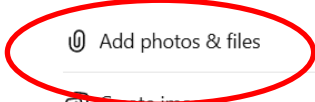
Analyze the Information

- LLM can help you analyse based on its own knowledge.

What are the current techniques and approaches to mitigate or reduce hallucinations in LLMs? What are the pros and cons of each approach?



- Or you can upload your resources and ask it to analyse for you

 Add photos & files

Create image

Thinking

Deep research

Study and learn

More

+ Ask anything

Synthesize Results

- **Synthesizing results** is an important step in the research process, especially after collecting and analyzing data. This helps:
 - Create a **comprehensive and systematic view of the research problem**: Helps researchers not only stop at "presenting data," but also draw out common meanings, prominent trends, or important relationships.
 - **Transform scattered data into meaningful information**: From many small analytical results (data tables, charts, interviews...), synthesis helps connect findings, clarifying consistency or contradictions in the data.

Synthesize Results

- LLM can help you synthesize result.

You are a professional data synthesis specialist. I am conducting research on "Hallucination phenomenon in LLMs". Please help me synthesize the information below and write a report including the following sections: Title, Introduction, Related Work, Analysis, and Conclusion.

+



- And help you export to Word or PDF

Limitations of ChatGPT

ChatGPT has the ability to synthesize documents but has some limitations:

- **Does not cite sources clearly:** ChatGPT does not indicate which file the synthesized content comes from, making it difficult to trace sources or cross-reference original information.
- **Prone to generating inaccurate information (hallucination):** ChatGPT may "infer" or "fill in the blanks" with information that seems reasonable but is not in the documents, causing content distortion.
- **Lacks ability to process logic between documents:** If there are conflicting documents, ChatGPT struggles to evaluate credibility to provide accurate or neutral synthesis.

Solution with NotebookLM

NotebookLM helps:

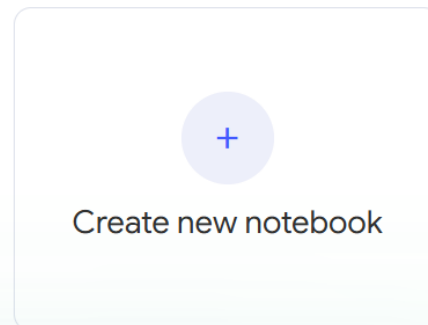
- **Display clear source citations:** NotebookLM **attaches citations to each synthesized text passage**, clearly indicating which section was taken from which document.
- **Reduce "hallucination" situations:** NotebookLM **limits the response scope to only the document set that you have uploaded**, helping to avoid "over-inference" or "fabricating information" that doesn't exist.
- **Easy document updating:** Simply add new documents to the working "notebook," NotebookLM will automatically add them to the synthesizing dataset, without needing to repeat the process from the beginning.

Practice with NotebookLM

Step 1: Go to <https://notebooklm.google.com/>

Welcome to NotebookLM

Recent notebooks



Try an example notebook

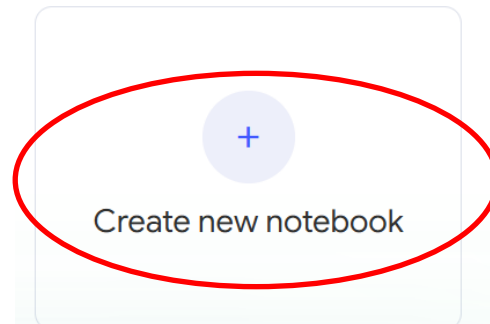
+ Create new notebook

Practice with NotebookLM

Step 2: Create a new Notebook

Welcome to NotebookLM

Recent notebooks

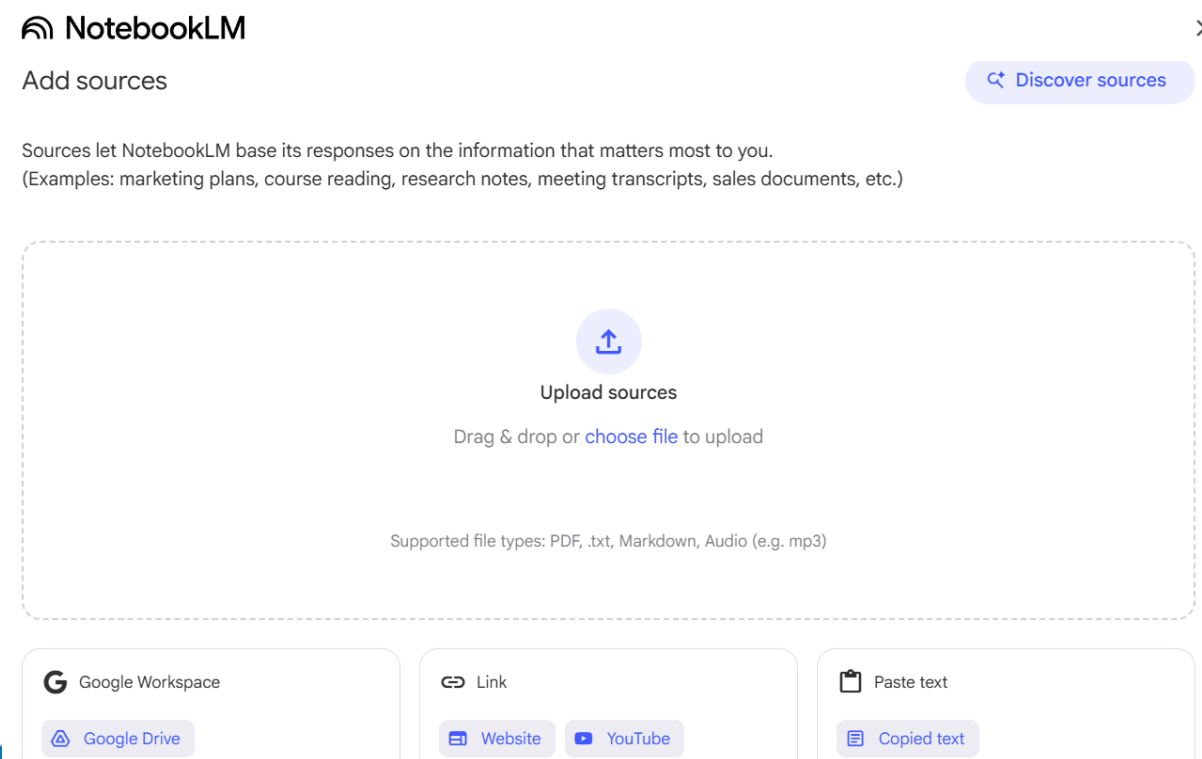


Try an example notebook

+ Create new notebook

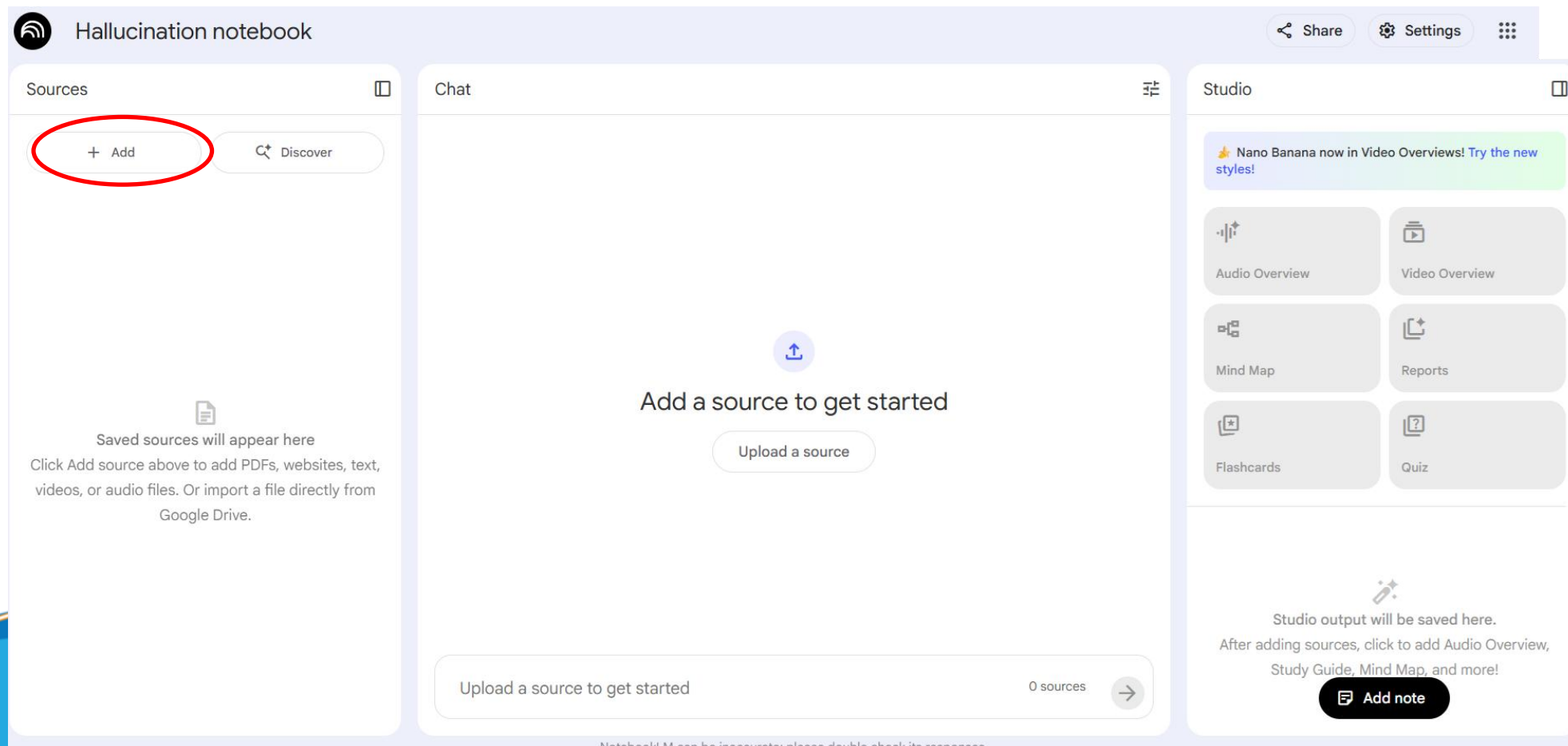
Practice with NotebookLM

Step 3: Close this dialog (Can open this dialog later). Name the Notebook (Ex: Hallucination)



Practice with NotebookLM

Step 4: Select "+Add" to add data source



The screenshot displays the NotebookLM interface, titled "Hallucination notebook". The interface is divided into three main panels: Sources, Chat, and Studio.

- Sources Panel:** Located on the left, it contains a red circle around the "+ Add" button. Below it, there is a "Discover" button. A message states: "Saved sources will appear here. Click Add source above to add PDFs, websites, text, videos, or audio files. Or import a file directly from Google Drive."
- Chat Panel:** The central panel, currently empty, with a large blue circular button in the center that says "Add a source to get started". Below this, there is an "Upload a source" button. At the bottom, a text input field says "Upload a source to get started" and a button indicates "0 sources" with a right arrow.
- Studio Panel:** Located on the right, it features a notification banner for "Nano Banana now in Video Overviews! Try the new styles!". Below this is a grid of six buttons: "Audio Overview", "Video Overview", "Mind Map", "Reports", "Flashcards", and "Quiz". At the bottom, a message states: "Studio output will be saved here. After adding sources, click to add Audio Overview, Study Guide, Mind Map, and more!" followed by an "Add note" button.

Practice with NotebookLM

Step 5: Upload the files downloaded (if desired)

Add sources

 Discover sources

Sources let NotebookLM base its responses on the information that matters most to you.
(Examples: marketing plans, course reading, research notes, meeting transcripts, sales documents, etc.)


Upload sources

Drag & drop or [choose file](#) to upload

Supported file types: PDF, .txt, Markdown, Audio (e.g. mp3)

 Google Workspace

 Google Drive

 Link

 Website

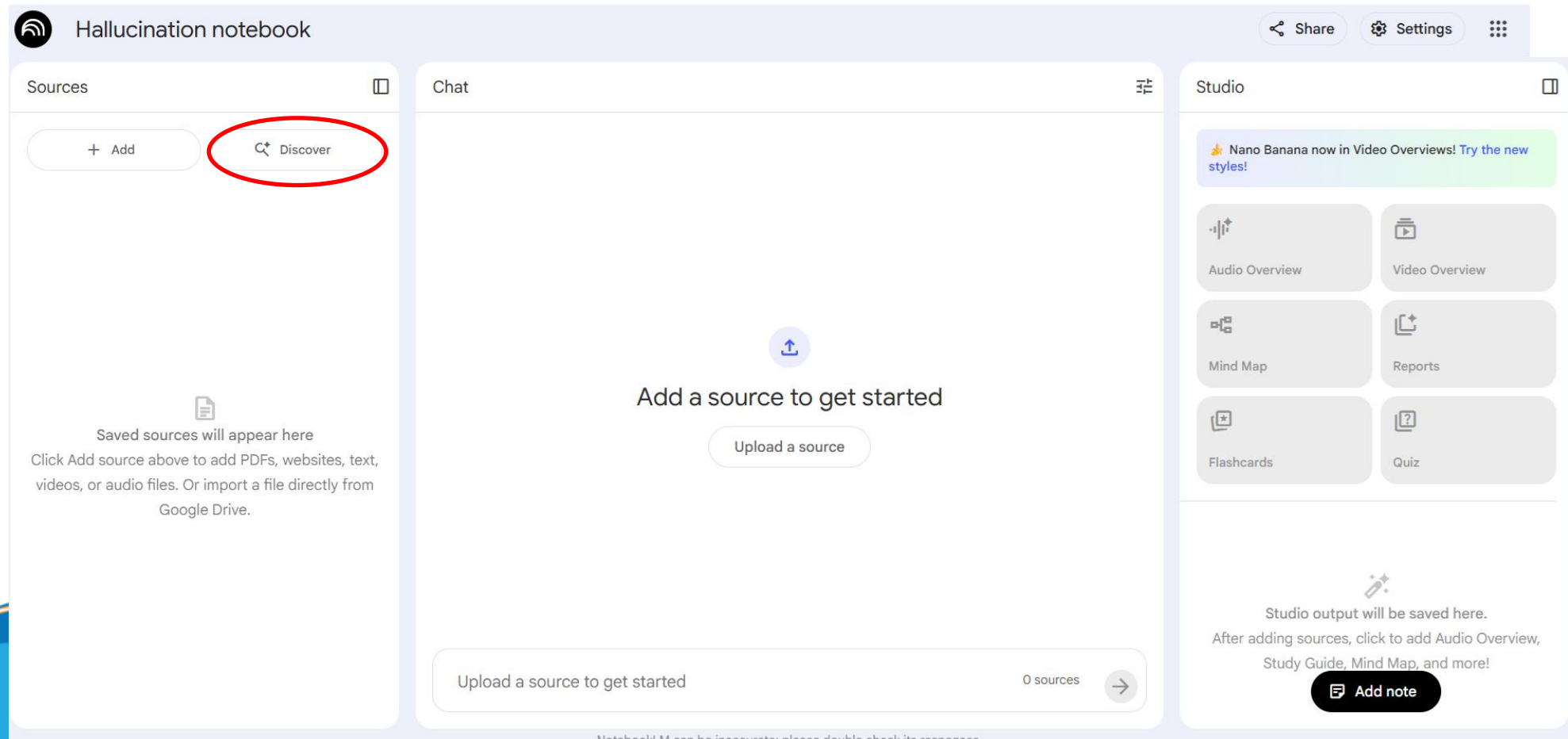
 YouTube

 Paste text

 Copied text

Practice with NotebookLM

Step 6: Select "Discover" for NotebookLM to automatically find data sources



Practice with NotebookLM

Step 7: Type the research topic in and press Submit

Discover sources



What are you interested in?

Hallucination Detection and Mitigation in LLMs

- Investigating methods to identify and reduce false or fabricated information generated by LLMs
- Developing frameworks for fact-checking and grounding responses in verifiable sources

Find sources from:

- ☒ Web
- ☐ Google Drive

 I'm feeling curious

Submit

Practice with NotebookLM

Step 7: Wait for NotebookLM to find news sources (10 sources), press Import

← Discover sources



These sources provide a balanced technical understanding of LLM hallucination, covering foundational causes, key mitigation strategies like RAG and fine-tuning, and the critical evaluation frameworks used for fact-checking.

Select all sources



Reducing Hallucinations and Trade-Offs in Responses in Generative AI Chatbots for Cance...



Examine RAG's practical performance and hallucination reduction rates in specific domains.



Fine-tuning and Utilization Methods of Domain-specific LLMs - arXiv



Review fine-tuning methods, like LoRA, for adapting LLMs and reducing fabrication.



GROUNDING LARGE LAN- GUAGE MODELS VIA DYNAMIC KNOWLEDGE ADAPT- ING OVER...



Detail the 'Chain-of-Knowledge' framework for dynamic knowledge grounding.



10 sources selected

Import

Practice with NotebookLM

Step 8: Wait for NotebookLM to process the data, then ask a question
(Ex: What is Hallucination?)

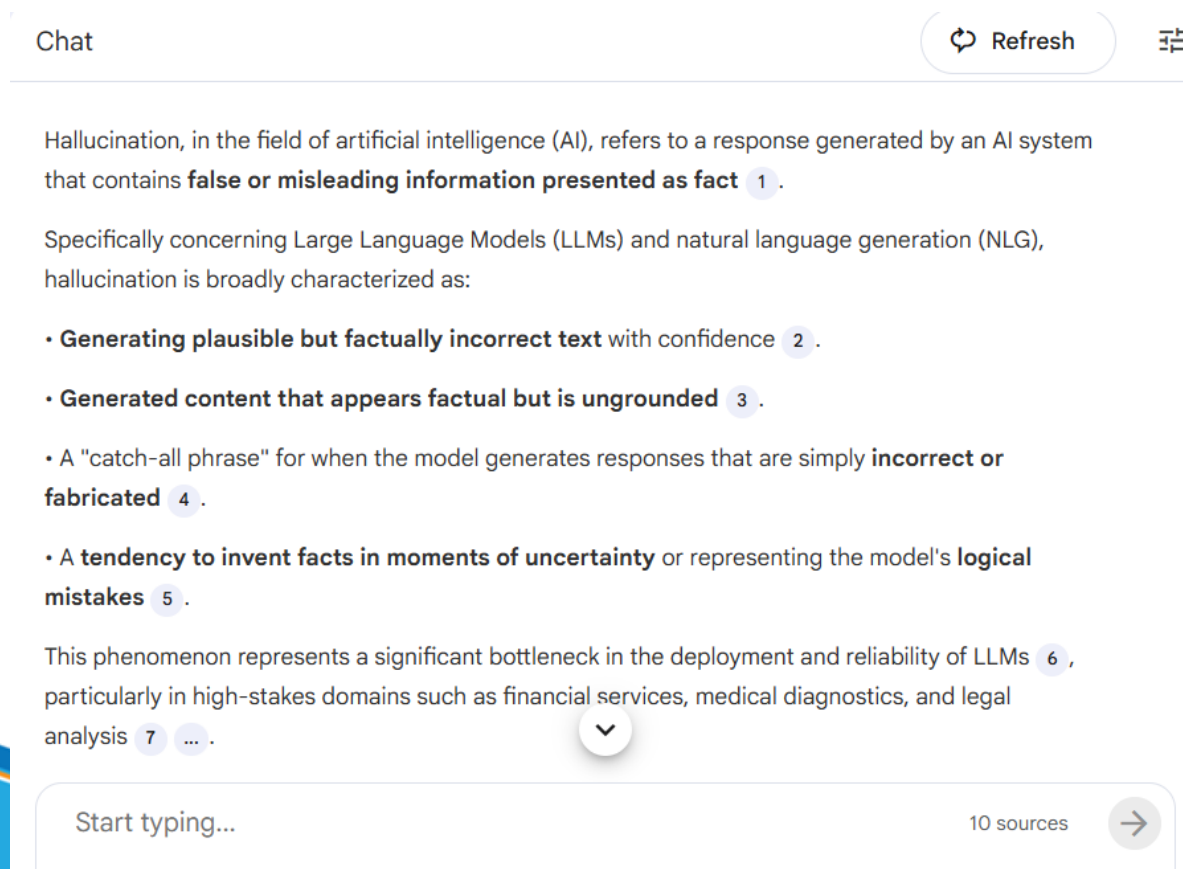
The screenshot displays the NotebookLM interface for a notebook titled "Hallucination notebook". The interface is divided into three main panels:

- Sources Panel (Left):** Contains a list of 10 sources related to Large Language Models (LLMs) and hallucinations. Each source has a checkbox to select it. The sources include:
 - 'Attention is all you need' coauthor says ...
 - Fine-tuning and Utilization Methods of ...
 - GROUNDING LARGE LANGUAGE MODEL...
 - Hallucination (artificial intelligence) - Wi...
 - LLM Hallucination Index - Kore.ai
 - OpenFactCheck: A Unified Framework f...
 - Prompt Engineering Best Practices You ...
 - Reducing Hallucinations and Trade-Offs...
 - Retrieval-augmented generation - Wikip...
 - Trustworthy Reasoning: Evaluating and ...
- Chat Panel (Center):** Shows a summary of the collected sources, focusing on challenges and enhancement methods for LLMs, particularly addressing AI hallucination. It mentions techniques like Retrieval-Augmented Generation (RAG) and Chain-of-Knowledge (CoK). Below the summary, there is a "Start typing..." input field (highlighted with a red circle) and a "Save to note" button. At the bottom, two example questions are shown: "What are the most effective methods for mitigating large language model hallucinations?" and "How does retrieval-augmented generation improve domain-specific factual accuracy?".
- Studio Panel (Right):** Offers various output formats for the generated content, including Audio Overview, Video Overview, Mind Map, Reports, Flashcards, and Quiz. A notification at the bottom states: "Studio output will be saved here. After adding sources, click to add Audio Overview, Study Guide, Mind Map, and more!" with an "Add note" button.

At the very bottom of the interface, a disclaimer reads: "NotebookLM can be inaccurate; please double check its responses."

Practice with NotebookLM

Step 9: Click on the citations (1, 8...) to see the source documents NotebookLM used to answer



Chat ↻ Refresh ≡

Hallucination, in the field of artificial intelligence (AI), refers to a response generated by an AI system that contains **false or misleading information presented as fact** ¹.

Specifically concerning Large Language Models (LLMs) and natural language generation (NLG), hallucination is broadly characterized as:

- **Generating plausible but factually incorrect text** with confidence ².
- **Generated content that appears factual but is ungrounded** ³.
- A "catch-all phrase" for when the model generates responses that are simply **incorrect or fabricated** ⁴.
- A **tendency to invent facts in moments of uncertainty** or representing the model's **logical mistakes** ⁵.

This phenomenon represents a significant bottleneck in the deployment and reliability of LLMs ⁶, particularly in high-stakes domains such as financial services, medical diagnostics, and legal analysis ⁷ ...

Start typing... 10 sources →

Practice with NotebookLM

Step 9: Click on the citations (1, 8...) to see the source documents
NotebookLM used to answer

Chat
Refresh

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1 INTRODUCTION

In recent years, large language models (LLMs) such as ChatGPT (OpenAI, 2023) have demonstrated impressive language generation capabilities (Cheng et al., 2023; Ding et al., 2023; Chen et al., 2024). However, one major challenge of LLMs lies in hallucination, which is their tendency to confidently generate plausible but factually incorrect texts (Ji et al., 2023; Zhao et al., 2023b). As shown in Figure 1, given a question, "What year was the Argentine actor who directed El Tio Disparate born?" which requires factual knowledge to answer, the most advanced LLMs often provide an incorrect answer. While LLMs have the remarkable capability to recall information from their training data, effectively updating

Practice with NotebookLM

There are other useful tool on the right side (Studio). It will generate various artifact from your source:

- Audio overview
- Video overview
- Mind Map
- Flashcards
- Report
- Quiz

👉 Nano Banana now in Video Overviews! [Try the new styles!](#)



Audio Overview



Video Overview



Mind Map



Reports



Flashcards



Quiz



Studio output will be saved here.

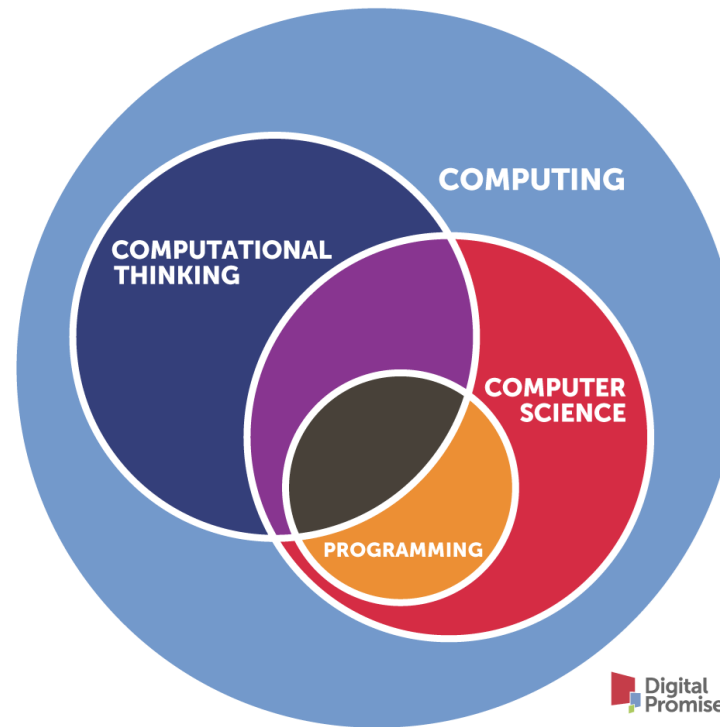
After adding sources, click to add Audio Overview,
Study Guide, Mind Map, and more!

 Add note

4. GenAI awareness and Human response

Introduction

What is computational thinking?



THANK YOU



Questions and Answers